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Commercializing Sustainable Technologies by Developing Attractive Value Propositions: The Case of Photovoltaic Panels

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Abstract

Despite their promise for tackling environmental problems, sustainable technologies are difficult to move from niche to mass markets. They require firms to develop innovative business models. This article focuses on how firms can develop their value propositions to increase the market attractiveness of sustainable technologies that are economically attractive but are difficult to commercialize. The results are based on an analysis of six German utilities that developed offers for power supply self-sufficiency using photovoltaic technology. The article proposes four generic tactics firms can use to market this technology—simplify, mimic, configure, and engage—and describes and discusses how each of these tactics can help firms deliver and capture value. The study also compares and contrasts how firms combine tactics to increase the attractiveness of their value propositions. The article contributes to the growing literature on business models for sustainability and proposes recommendations to help sustainable innovation overcome customer inertia.

Keywords

sustainable technology, business model, value proposition, utilities, photovoltaic

Introduction

Sustainable technologies often fail to move from niche to mass markets (REN21, 2016) despite their promise for tackling environmental problems (Hockerts & Wüstenhagen, 2010; Johnson & Suskewicz, 2009). Reaching sustainability targets such as ambitious reductions in greenhouse gas emissions requires a sustainable transformation of mass markets (Schaltegger, Lüdeke-Freund, & Hansen, 2016).

Previous studies have identified various explanations for this failure. Notably, technological and institutional lock-ins may reinforce incumbent technologies at the expense of new and more

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sustainable ones (Foxon, 2002). Environmental and innovation policies may help overcome such lock-ins by establishing sufficient demand (Foxon & Pearson, 2008). In particular, in many countries renewable support mechanisms such as feed-in-tariffs have created markets for high-cost renewable energy technologies and helped overcome the technological fossil fuel lock-in in the electricity sector (del Río & Bleda, 2012; Unruh, 2000, 2002).[1]

At the firm level, sustainable technologies may also involve technological shifts which firms find difficult to manage (Tongur & Engwall, 2014). Moreover, sustainable technologies can be disruptive and have been found to underperform established competing technologies (Christensen, 1997). Because sustainable technologies typically require customers to change their preferences, they often lack market attractiveness (Johnson & Suskewicz, 2009) and oblige firms to translate the characteristics of the technology into (new) ways to create economic value (Bocken, Short, Rana, & Evans, 2014; Boons & Lüdeke-Freund, 2013).

While the literature has studied the impact of policies on sustainable technology diffusion extensively (see, for instance, Gillingham & Palmer, 2014; Hoppmann, Huenteler, & Girod, 2014; Negro, Alkemade, & Hekkert, 2012; Veugelers, 2012), a complementary body of literature advocates studying how firms innovate their business models to make sustainable but possibly inferior technologies attractive (Chesbrough & Rosenbloom, 2002; Drury et al., 2012; Sousa-Zomer & Cauchick Miguel, 2016; Yu & Hang, 2010). Previous research has identified several business model archetypes which overcome technological and institutional barriers (see, for instance, Bohnsack, Pinkse, & Kolk, 2014; Sauter & Watson, 2007). The article by Bohnsack and Pinkse (2017) is one of the few studies that goes beyond proposing archetypes to explore in detail how these business models are configured. They identified three tactics that firms can use to reconfigure their value proposition for disruptive technologies and increase the market attractiveness of technologies that are immature and costlier than incumbent ones.

Sustainable technologies can be difficult to diffuse, even when they are mature and as costly as (or cheaper than) incumbent technologies. For instance, in many regions, electricity generated by photovoltaic modules (PV) is economically attractive (Auer & Lettner, 2012), yet it remains difficult to introduce on the mass market (Karakaya, Hidalgo, & Nuur, 2015). This suggests that costs and technological maturity alone may be insufficient to explain sustainable technologies' lack of market success. In this article, we argue that the tactics identified by Bohnsack and Pinkse (2017) may not be sufficient to support the development of compelling value propositions for sustainable technologies.

The objective of this article is therefore to identify tactics to increase the attractiveness of sustainable technologies in the market, discuss how these tactics impact business models, and compare how firms combine them to develop compelling value propositions.

We achieve these objectives by analyzing how firms develop their value propositions to commercialize PV technology. Our analysis focuses on utilities that have developed business models for power supply self-sufficiency based on PV technology for residential consumers in Germany, a country historically known for its policies promoting the diffusion of PV technology (European Commission, 2014).[2]

This article makes three key contributions to the literature. First, it identifies and contrasts four tactics—simplify, mimic, configure, and engage—that utilities can employ to develop compelling value propositions for sustainable technologies and discusses the impact of those tactics on business models. Second, by comparing how firms combine these tactics, this article discusses trends and challenges facing the diffusion of PV to the mass market. And third, by deriving recommendations for overcoming customer inertia regarding sustainable technologies, this article also contributes to the more general literature on business models for sustainability (Schaltegger et al., 2016).

The remainder of the article is organized as follows. Section 2 introduces existing research on business models for commercializing sustainable technologies. This includes research on

business model archetypes and tactics for reconfiguring value propositions for sustainable technologies to increase their market attractiveness. Section 3 also explains why PV technology and especially the market for power supply self-sufficiency is particularly suitable for analyzing how firms can develop attractive value propositions for sustainable technologies that have limited market attractiveness despite being technologically mature and economically competitive. Section 3 explains the research method. Section 4 describes how the firms in the sample develop their value propositions to overcome barriers to PV technology adoption. Section 5 discusses the findings and proposes four tactics that firms can employ to develop attractive value propositions. Section 6 proposes a cross-case analysis. Comparing and contrasting the combinations of tactics firms use, it discusses how these tactics relate to company background. In addition, Section 6 derives insights about evolving customer segments and explains why firms targeting tenants use limited combinations of tactics. The final Section 7 concludes, provides general recommendations for developing a compelling value proposition for sustainable technologies and suggests avenue for future research.

Literature Review

Sustainable and potentially disruptive technologies frequently lack market attractiveness: they are often expensive, underperform compared with incumbent technology (Christensen, 1997) and require customers to change their preferences (Johnson & Suskewicz, 2009). A growing body of literature suggests that one way to make these technologies attractive is to commercialize them through innovative business models (Bocken et al., 2014; Boons & Lüdeke-Freund, 2013; Chesbrough & Rosenbloom, 2002; Yu & Hang, 2010). According to Teece (2009), a business model reflects a management "hypothesis about what customers want, and how an enterprise can best meet those needs, and get paid for doing so" (p. 24). It invites managers to think about how they can create and deliver value to their customers, and how they can capture part of this value for themselves (Teece, 2010).

In the following sections, we introduce business model archetypes for sustainable technology. We discuss tactics identified in prior literature to increase the attractiveness of disruptive technologies and argue that these tactics are not sufficient to explain how firms can commercialize sustainable technologies.

Business Model Archetypes for Commercializing Sustainable Technology

Previous research has aimed at identifying overarching business model archetypes that are or can be used to commercialize sustainable technologies. These technologies include innovations such as electric vehicles (EVs), solutions designed to develop a circular economy, and distributed energy-generation technologies. First, building on the work of Tukker (2004), Kley, Lerch, and Dallinger (2011) argue that business models for EVs may be product-oriented or service-oriented. If they are service-oriented, they may then be use-oriented or results-oriented. Focusing on product-oriented business models, Bohnsack et al. (2014) categorize four business model archetypes depending on whether an EV is specialized or multipurpose and addresses the luxury or economy market segment. Second, recent research also identifies various business model strategies for the circular economy (Bocken, de Pauw, Bakker, & van der Grinten, 2016; Bocken et al., 2014; Bocken & Short, 2016). This includes among others business models that deliver functionality rather than ownership (also called product service systems; see Sousa-Zomer & Cauchick Miguel, 2016; Wells, 2016), maximize resource and energy efficiency, and create value from waste or encourage self-sufficiency. Finally, the extant literature classifies four generic business models for utilities seeking to incorporate distributed generation in their portfolios. First, utilities can sell distributed electricity systems as turnkey solutions (Sauter & Watson, 2007; Schoettl &

Lehman-Ortega, 2011; Strupeit & Palm, 2016). Second, they may adopt a third-party business model and build, own, operate, and lease distributed electricity systems to a building's owner or residents (Schoettl & Lehman-Ortega, 2011; Strupeit & Palm, 2016). Third, utilities can provide extra services such as financing for or maintenance of distributed electricity systems or develop offers for integrating small-scale producers into the electricity market as part of a virtual power plant system (Schoettl & Lehman-Ortega, 2011). Fourth, they can develop a utility-sponsored community shares business model which offers multiple users the possibility of purchasing a portion of their electricity from a renewable energy facility located off-site (Asmus, 2008; Funkhouser, Blackburn, Magee, & Rai, 2015; Sauter & Watson, 2007).

Developing Attractive Value Propositions for Immature Disruptive Technology

While the existing literature proposes generic business model archetypes, few papers deconstruct these business models to explain how exactly they are configured (Charter, Gray, Clark, & Woolman, 2008; Schaltegger, Lüdeke-Freund, & Hansen, 2012; Tukker & Tischner, 2006; Wells, 2008). One exception is Bohnsack and Pinkse (2017), who offer a detailed analysis of how firms that commercialize a disruptive, costly, and immature technology can create value for their customers. More specifically, they study how car manufacturers reconfigure their value propositions to increase the market attractiveness of EVs.

Accordingly, the value proposition comprises a bundle of value elements (Lindič & Marques da Silva, 2011) and customers choose products or services that have the most attractive bundles of value elements (Anderson, Narus, & Van Rossum, 2006). By reconfiguring their value propositions, firms can propose a range of combinations of value elements and thereby reach a wider market segment. Bohnsack and Pinkse (2017) propose three tactics for doing so. First, firms can look for ways to compensate for the inferiorities of the technology to reach value parity with incumbent technology. Car manufacturers can, for instance, propose substitute cars to compensate for the short driving ranges of EVs. Second, firms can use enhancing tactics and add value elements which are not covered by incumbent technology. For example, some firms propose car-sharing services to customers who acquire EVs, thereby accessing untapped customer value. Third, firms can adopt a coupling strategy and build on customer value for a complementary product developed by firms in other sectors to offer combined solutions. For instance, various car manufacturers in the United States have teamed up with solar system providers to allow customers to recharge their cars with home-generated renewable electricity. Bohnsack and Pinkse (2017) argue that reconfiguring the value proposition can be a timelier manner to increase the market attractiveness of a disruptive technology that is immature and costly than focusing on technological innovation.

Developing Attractive Value Propositions for Sustainable Technology

We argue that the tactics proposed by Bohnsack and Pinkse (2017) are not sufficient to explain how firms can commercialize sustainable technologies. Indeed, many sustainable technologies are difficult to move from niche to mass markets even when they are mature and economically attractive to end users. Energy efficiency measures, for instance, are often not implemented by firms even though they appear to be economically attractive and easy to implement (Fleiter, Schleich, & Ravivanpong, 2012; Olsthoorn, Schleich, & Hirzel, 2017). This situation is known in the literature as the energy efficiency paradox (Jaffe & Stavins, 1994). Other sustainable technologies such as PV technology (Karakaya et al., 2015), solar boilers (Yang, 2010), eco-efficient services (Anttonen, Halme, Houtbeckers, & Nurkka, 2013), sustainable technologies for water treatment (Mankad & Tapsuwan, 2011; Sousa-Zomer & Cauchick Miguel, 2016), or green buildings and green roofs (Hoffman & Henn, 2008; Zhang, Shen, Tam, & Lee, 2012) face a similar problem.

Sustainable technologies often face barriers that are not technological but that relate to organizational, social, or even cultural issues instead. Potential adopters often perceive sustainable technologies as complex (Luthra, Kumar, Garg, & Haleem, 2015; Strupeit & Palm, 2016) or risky (Anttonen et al., 2013; Mankad & Tapsuwan, 2011; Olsthoorn et al., 2017; Sousa-Zomer & Cauchick Miguel, 2016), as involving sacrifices (Hoffman & Henn, 2008; Howard-Grenville & Hoffman, 2003), or requiring changes in organizational routines (Anttonen et al., 2013; Zhang et al., 2012). Besides, sustainable technologies may raise little interest from customers (Fischer, 2008) or are not considered to be strategically important (Cooremans, 2011). These barriers suggest that costs and technological maturity alone are not always sufficient to explain sustainable technologies' lack of market success. This means that the tactics developed by Bohnsack and Pinkse (2017) may not be sufficient for broadening the market segments of sustainable technologies. Building on this work, our ambitions for this article are threefold. First, we aim to identify tactics that can be used to increase the market attractiveness of sustainable technologies that are mature and cost-efficient but still largely fail to reach the mass market. Second, we discuss the impact these tactics have on business models. Third, we propose a cross-case comparison to compare and contrast the combinations of tactics used by the firms studied.

Self-Sufficiency as a Suitable Context for Identifying Tactics for Developing Attractive Value Propositions for Sustainable Technology

We argue that PV technology and especially the market for power supply self-sufficiency is well suited to analyze how firms can develop attractive value propositions for commercializing a sustainable, mature, and economical yet not very attractive technology.

PV technology holds the promise of helping to decarbonize the electricity sector. In recent decades, the technology has undergone various waves of technological development and the technology is said to have reached maturity (U.S. Department of Energy, 2012). Moreover, technological development has also dramatically reduced the cost of PV panels. To suggest an order of magnitude, between 2008 and 2014, the cost of PV installations fell by about 70% (European Commission, 2014). In some regions where electricity prices are relatively high, including Germany, Spain, and the Netherlands, PV has already reached grid parity (aka. socket parity; Auer & Lettner, 2012).[3] This enables the price of PV-generated electricity for end users to compete with that of electricity generated by conventional sources of energy. Globally, more and more PV panels are being installed and in 2014 the EU exceeded its 2020 objectives of having 84.4 GW of PV panels in place. The market for self-sufficiency in power supply is seen as a way to speed up the diffusion of PV panels and place consumers at the heart of the energy transition (Schleicher-Tappeser, 2012). Besides, lowering energy bills and helping grid management (Moshövel et al., 2015), self-sufficiency could help finance the transition of the electricity sector (e.g., Blomberg New Energy Finance, 2015). A recent study estimated that about half of all EU households may be able to become prosumers, that is, they would both produce and consume electricity (CE Delft, 2016). In June 2018, the EU amended the Renewable Energy Directive (Directive 2009/28/EC) in order to ensure a more favorable regulatory environment for household self-consumption and further stimulate this market.

Nevertheless, in most countries, solar power still represents only a small percentage of electricity generation and prosumers represent a very small minority of citizens (IEA-RETD, 2016). In other words, even though PV panels are a mature and (in some instances) cost-efficient technology, they have proved difficult to move from a niche to a mass market.

Scholars have identified several barriers PV technologies face. First, transaction costs for customers are high: customers are not familiar with PV technology (Yang, 2010), and many feel that the technology and the processes surrounding it are too complex. Choosing a PV system, for

instance, can be difficult because many options exist that are difficult to compare in terms of price and associated services (e.g., maintenance, monitoring, and warranty; Rosoff & Sinclair, 2009; Strupeit & Palm, 2016). Installation can also be time-consuming and require some coordination. Even though early adopters are generally more willing to spend time to appropriate this technology (Moore, 1999), mass-market customers demand well-defined options that can be easily implemented (Schleicher-Tappeser, 2012). Moreover, in the past, customers have raised concerns about the durability, efficiency, and reliability of PV systems (Karakaya et al., 2015; Rosoff & Sinclair, 2009). In addition, high upfront and maintenance costs are also partly responsible for the inertia (Karakaya et al., 2015; Rosoff & Sinclair, 2009; Shih & Chou, 2011). Finally, the deployment of PV technology suffers because electricity is a commodity that cannot be differentiated and to which few consumers are emotionally bound (Fischer, 2008). These issues are exacerbated by the fact that for many years consumers were excluded from decision-making processes in an electricity system that was centrally planned by state-owned, vertically integrated companies. Consumers interacted with electricity only when paying their bills or experiencing technical problems. Most consumers still interact with electricity in that way and the industry often describes electricity as a "low-interest product" (Fischer, 2008).

Research Method

Self-Sufficiency in Electricity Consumption: The German Context

Business models depend heavily on local context (Strupeit & Palm, 2016), so we chose to focus on a single market (Germany) where self-consumption is economically attractive because of high end use prices (Henkel & Lenck, 2013) and has been promoted politically via various support schemes such as a bonus for self-consumed electricity or exempting self-consumed electricity from certain levies (European Commission, 2014). Germany is considered a pioneer in the field of energy, climate and innovation policy, and through the "Energiewende," that is, the energy transition or energy turnaround, the country decided to move away from fossil and nuclear energy sources. In particular, in the wake of the Fukushima nuclear accident in 2011, the German government decided that renewable energy sources should account for 80% of the country's electricity production by 2050. The Energiewende is facilitated by framework conditions such as the Renewable Energies Act (Erneuerbare-Energien-Gesetz [EEG]). EEG prioritizes feed-in of electricity from renewable energy sources, guarantees fixed remuneration per kWh produced over 20 years, and has promoted renewable energies from being niche products to becoming an important part of the German energy mix. In 2011, renewable energy represented 20% of Germany's electricity production (BMUB & BMWi, 2011); by 2017, this share had increased to about 38% (Fraunhofer Institute for Solar Energy Systems ISE, 2017).

In Germany, the electricity price for private households includes various taxes and levies, and to promote self-sufficiency. Some of these added costs are fully or partially waived if the electricity is consumed where it is produced. These exemptions are crucial to create a market for self-sufficiency and delineate it. For instance, since the EEG partially waives the levy for PV installations with a capacity below 10 kWp and self-consumption of less than 100 MWh p.a., firms' market penetration strategies focuses on installations meeting these qualifications. The German regulatory context partly explains why the increase in renewable electricity production has been accompanied by an increase in self-consumption. Prior to 2011, hardly any electricity was self-consumed; in 2012, after the regulatory framework changed, 0.5% of net electricity consumption in the household sector was provided by self-produced power (Henkel & Lenck, 2013). With the rise in the retail price of electricity and falling costs of battery storage systems, battery-coupled PV systems may meet up to 30% of electricity consumption in the household sector by 2035 (Prognos, 2016).[4]

We also chose to limit our analysis to utilities. German utilities own and operate only a small percentage—11.9% in 2013—of German renewable energy-generation capacity (Trendresearch, 2013). Small actors, that is, individuals, farmers, and project developers, own most of the currently installed renewable energy-generation capacity (Wassermann, Reeg, & Nienhaus, 2015). In other words, to prevent a further decline in market share and limit potential revenue losses German utilities also need to find ways to capture part of the growing market for electricity self-sufficiency.

Case Introduction

By the end of 2015, only a few utilities had begun to pursue a business model designed to achieve self-sufficiency in power supply. Among the 1,300 active electricity providers in Germany, fewer than 20 had offerings specifically related to self-sufficiency. Data were collected for six firms. The selection criteria applied required firms to target residential end users and not private firms and the offerings had to be related to PV panels. Firms may target both homeowners and tenants. The latter offerings are known in the industry as Mieterstrom offerings ("power for tenants") and are often developed by firms to overcome the "justice gap" between tenants and owners. Unlike tenants, owners have long been able to produce and consume the power generated by systems on their properties, thereby benefiting from financial support granted under the EEG. We also selected various types of firms to reflect the variety of energy providers present in Germany.

Case 1: Beegy. Beegy is a start-up founded in November 2014 by a consortium of firms including a regional energy provider (MVV Energie), a renewable energy service provider (BayWa Renewable Energies), a manufacturer of electrical heating appliances (Glen Dimplex Deutschland) and an energy IoT Platform developer (GreenCom Networks). As part of its core business, Beegy developed a self-sufficiency PV offer called Beegy Solar targeting homeowners. Beegy organizes the planning and installation of PV systems. Beegy proposes a contract for a fee of 149€/year that includes the following advantages. First, it guarantees 50% cost savings on the electricity bill of the previous year. This is achieved through free consumption of self-produced electricity and feed-in-tariff remuneration. Second, it proposes access to a learning PV plant. Beegy developed algorithm-based software that analyzes the predicted solar production, learns from user behavior, and derives personalized tips to increase self-consumption. Finally, the offer includes remote technical monitoring to anticipate problems and avoid manual interventions.

Case 2: DZ-4. DZ-4 is a start-up founded in September 2011. The firm offers a full service package for customized leasing of PV and possibly storage systems as part of its core business. This means that DZ-4 cares for a household's complete power supply and they have a contract with a green electricity provider to deliver the residual power. DZ-4 also installs, operates, monitors, and maintains the PV (and possible storage) system. Finally, customers do not have to bear any initial investment costs. DZ-4 proposes two offerings: (a) DZ-easy, which includes only PV panels on the roof and can cover about 1/3 of a customer's electricity demand; (b) DZ-Autark, which also includes a storage system so a customer can cover about 2/3 of his or her electricity demand. In 2018, the firm had 30 employees and four-digit customers (Noack, 2018). The incumbent utility EnBW owns 15% of DZ-4.

Case 3: RWE. RWE is a large incumbent utility operating worldwide. The company launched in April 2015 a campaign called "solar offerings." It sold turnkey and storage-ready solar systems with an 11-year warranty, 5-year insurance coverage, and 2-year monitoring service to homeowners. If customers covered their residual power needs with electricity delivered by RWE, the monitoring period is extended indefinitely. RWE explained selecting only proprietary brands

from Germany, standing for high quality for the delivery of the PV modules, power inverters, and frame. Customers had the possibility to combine this offering with the purchase of a battery storages system. This campaign was meant as a gateway to an emerging market and a way to retain their customer base. In April 2016, RWE created the subsidiary Innogy SE that combined all its activities linked to renewable energy. Innogy is planned to be sold to the German utility E.ON that will own a little more than 85% of the shares.

Case 4: HEG. HEG is an energy cooperative created in 2010 in the city of Heidelberg as a result of a student initiative. The aim of the cooperative is to realize renewable energy projects in Heidelberg (Heidelberger Energiegenossenschaft, 2018b). In 2012, they launched a new initiative—a "Mieterstrom" offering—with the aim to allow tenants to consume PV electricity produced on the roof of their multifamily rental houses at lower prices relative to conventional power. This initiative was experimental and included seven apartment houses in the "Neue Heimat" area in Nußloch, a small municipality near Heidelberg (Heidelberger Energiegenossenschaft, 2018a). Tenants started to self-consume electricity in December 2014. To benefit from their offer, tenants only have to change their electricity supplier and enter a contract with HEG. They pay a base price, a price for locally produced electricity, and a price for residual power. HEG takes care of buying, installing, operating, monitoring, and maintaining the PV system. They are also in charge of the metering. Since the completion of the experiment, the cooperative has organized multiple workshops to share its experiences of this complex type of offering.

Case 5: Engynious. Engynious is a provider of decentralized renewable energy solutions. This SME created in 2001 launched a Mieterstrom offering called "solar plus power" in 2013 with the aim to allow tenant to self-consume PV electricity produced on the roof of their multifamily rental houses. This offering is directly linked with the core activity of Engynious and the multifamily segment is seen as of strategic importance by the firm (Engynious Energy Solutions, 2018). The first project took place in a municipality called Wolfen in East Germany. To participate, tenants only need to change their electricity supplier and enter a contract with Engynious. They pay a mixed price encompassing the locally produced electricity and the residual electricity coming from the grid. Engynious invests in the PV system and takes care of installation, operation, monitoring, maintenance, and metering.

Case 6: Lichtblick. Lichtblick is a renewables utility created in 1998. This SME launched in 2014 the "ZuhausStrom Mieterstrom" offering to allow plant operators (e.g., property owners or cooperatives) to supply tenants with electricity produced in a CHP or a PV system installed at multifamily rental houses at lower prices relative to conventional power. The ZuhausStrom offering represents a gateway to an emerging market for the firm. LichtBlick is responsible for the full supply of electricity to tenants. They pay a base price and a special ZuhausStrom price, which is a mix of locally produced electricity and residual green power. In this offering, plant operators take care of buying, installing, and maintaining the CHP or PV system. Their first project was realized in Berlin and concerned 3,000 tenants in 50 buildings.

Data Collection and Analysis

The results presented are based on an exploratory qualitative research methodology. Internet searches and personal exchanges with energy market experts identified the utilities with offerings specifically targeting self-sufficiency. Data were collected through desk research and six semi-structured interviews with utility representatives including managing directors, sales directors, innovation managers, and project managers (see Table 1, for details). Interviews were conducted

Table 1. Overview of the Interviews Conducted.

Company	Person interviewed	Type, date of contact
MVV Energie AG	Product and innovation manager, in charge of the development of BEEGY GmbH	Personal interview, 07.08.2015
BEEGY GmbH	Director of sales B2C	Phone interview, 07.20.2015
DZ-4 GmbH	Head of procurement, plant engineering, and operation	Phone interview, 08.19.2015
Engynious Deutschland GmbH	Managing director	Phone interview, 09.29.2015
LichtBlick SE	Project manager "ZuhauseStrom"	Phone interview, 10.19.2015
RWE Vertrieb AG	Responsible for sales to private customers and for special customers and grid processes	Phone interview, 11.06.2015

between June and November 2015 either face-to-face or via phone calls, and lasted between 30 and 45 minutes. Detailed notes were taken during the interviews. The interviews enabled us to gather firsthand information about the driving forces behind the development of the offerings, about the business models associated with them, and about the strength and weaknesses of these business models. We also participated in a workshop in September 2015, which was organized by an energy cooperative, the HEG Heidelberger Energiegenossenschaft. The workshop brought together expert stakeholders from utilities, nonprofit organizations, and universities. It aimed at presenting HEG's self-sufficiency business model for Mieterstrom offerings (targeting tenants) and discussing firsthand insights into the current state of the art of these types of offerings and the main challenges ahead. Detailed notes about the content of the presentation and the following discussions with the audience were taken in order to identify the main element of HEG's business model as well as key challenges surrounding this type of business model.

We started data analysis by detailing each business model and composed short summaries describing how each business model creates, delivers, and captures value. This exercise revealed that the most interesting differences between the cases related to how the firms aimed to create value. Firms tried to make their offerings attractive to customers by developing unique value proposition configurations. These observations are in line with those of Bohnsack and Pinkse (2017), who argued that firms use a range of tactics to increase the attractiveness of their products or services. We then formulated the value proposition for each of the sample firms and specifically tried to identify how these firms developed their value propositions to overcome barriers previously identified in the literature: high upfront costs, the complexity of the technology, lack of trust in the technology, and the fact that electricity is a low-interest product. We then tried to relate the strategies used to develop the value propositions to the tactics earlier identified by Bohnsack and Pinkse (2017), namely compensating, enhancing, and coupling. We found various compensating and enhancing tactics but no coupling tactics. More important, the empirical data revealed significant differences between the types of compensating and enhancing tactics used by the firms. We further refined these tactics to propose subgroups of tactics for developing attractive value propositions.

To identify the additional impacts of each tactic on the respective business models, we analyzed how firms deliver and capture value. We specifically tracked the value networks put in place by firms and the types of competencies needed to deliver the offers. We also reconstructed the firms' cost and revenue models by identifying the sources of revenue and determining whether they are recurring or nonrecurring, and then assessing the extent to which the business models require substantial assets (whether they are asset heavy or asset light).

Results: Configuring Value Proposition Around Secondary Benefits

Analyzing the value propositions of the case study firms revealed that each has a core value proposition and also offers secondary benefits meant to overcome one or more of the challenges involved in commercializing PV technology. The core value propositions of firms targeting homeowners (RWE, Beegy, and DZ-4) are very similar and consist in providing access to cheap renewable electricity and increased independence from the grid. Similarly, firms proposing Mieterstrom offerings for multifamily houses (Engynious, HEG, and Lichtblick) also have very similar core value propositions. These value propositions are two-sided: one is targeting tenants and one is targeting real estate owners. For tenants, they offer cheap access to renewable electricity produced partially on-site. For their part, real estate owners benefit from increased real estate value and can offer locally produced electricity to tenants.

The remainder of this section describes in greater detail the secondary benefits proposed by each of the firms in their attempts to overcome one or more of the barriers facing PV technology. The presentation of results is organized around the main challenges identified in the literature, namely high upfront costs, perceived complexity, lack of trust, and the need to increase customer interest in the technology.

High Upfront Costs

Half the sample firms developed strategies for overcoming the high upfront cost of PV technology. Beegy, RWE, and LichtBlick did not develop value propositions that minimize the initial investment needed by their customers. In their offerings, customers own the PV (and possibly storage) system and have to bear the upfront costs.

HEG, DZ-4, and Engynious do however offer to minimize the financial burden on their customers. HEG and Engynious invest in PV systems and sell the renewable electricity produced by the PV panels to participating tenants living in the buildings where a project is carried out. This takes the financial risk of the project away from real estate owners, who may therefore be more willing to have PV panels installed on the roofs of their properties. For the firms, however, the risk is higher, as they not only have to secure the investment in the PV system but also because their revenue depends heavily on the connection rate, namely the percentage of tenants who decide to switch to locally supplied electricity.

DZ-4 adopts a slightly different approach and leases PV panels to customers, who have to pay only a monthly fee. The initial contract is signed for 10 years and after this period customers can either purchase the system or continue leasing it. This proposition is capital intensive for DZ-4, whose development is tightly bound to securing funds from external investors.

Perceived Complexity of the Technology

All the firms studied adopted strategies for reducing the perceived complexity of the technology. First, PV panels installed on a roof can only partially meet a household's demand for electricity (no electricity is produced at night, for instance). The rate of self-sufficiency can be increased by investing in a storage system. But unless one invests in large PV and storage capacity, residual power must be purchased from the grid. This means that, under business-as-usual circumstances, customers need two contracts: one for the PV (and storage) system and one for the residual power. The first strategy employed by suppliers to reduce complexity refers to meeting customers' total demand for electricity with PV technology and residual power. Except for Beegy and RWE, for which it is optional, all of the firms in the study offer this solution to their customers. This enables them to lower customer transaction costs by relieving them of having to arrange two

separate contracts. Utilities may supply the remaining electricity themselves and benefit from additional revenues (Lichtblick; HEG) or buy it from a partnering (often green) energy supplier (DZ-4, Engynious).

Second, sample firms offered to minimize customer efforts during installation and operation of the systems. All but LichtBlick organize the planning and installation of their PV (and storage) systems, which they accomplish either internally or by externalizing to regional specialists who are already present throughout the country. Moreover, once a system is operational, all the firms take care of monitoring the systems. This also enables utilities to interact regularly with their customers (homeowners for Beegy, DZ-4 and RWE and real estate owners for HEG, Engynious, and LichtBlick). For RWE, the offering is limited to a 2-year monitoring service which can be extended indefinitely if residual power is purchased from RWE. This strategy also increases the likelihood of cross-selling, which in turn increases firm profits and strengthens customer relations.

Finally, the three firms proposing Mieterstom offerings for multifamily houses also include administration (i.e., billing), customer service, and communication with distribution system operators (DSOs). The billing is quite complicated in these projects. Firms have to measure who consumes how much and when. They also have to trace how much locally produced electricity is consumed and how much has to be supplied by other sources or sold to the grid. Moreover, interaction with the DSOs is also key and the firms under study emphasized that good collaboration with a DSO is crucial to ensuring the success of a project. DSOs are usually responsible for the metering of electricity consumption. If the metering is performed by another firm, the DSO will have to log off its own meters, and arrangements have to be made to transmit the DSO data on the electricity produced and consumed. By offering these services, firms minimize the burden on real estate owners who do not have to bear additional responsibilities if a PV system is installed on one of their rooftops.

Lack of Trust in the Technology

All the firms in our sample targeting homeowners try to manage the common perception that the technology is risky. First, RWE developed a partnership with the insurance company AXA S.A., enabling them to offer an 11-year warranty on the PV installation, and 5 years of all-risks insurance. Moreover, RWE also note that only proprietary brands from Germany that represent high quality are chosen for the delivery of the PV modules. RWE also leverage their well-known brand and long company history to further strengthen the trust of their customers in the products they are commercializing. Beegy and DZ-4, as small start-ups, have to develop other strategies to convince their customers of the seriousness of their offers. DZ-4 does this by proposing tariffs to its customers that are fixed for 10 years. Given that electricity prices have been regularly increasing in Germany, their objective is to give customers long-term visibility and predictability over their electricity prices. Moreover, by internalizing the operation and maintenance costs of its systems, the firm removes all technical risks from customers. Finally, for Beegy the situation is unique given that they sell a service that enables customers to save 50% on their electricity bills. Customers achieve this by self-consuming the electricity produced by their PV panels and by receiving extra feed-in remuneration for the surplus electricity injected into the grid. To convince customers that the service can be delivered, Beegy guarantees that it will refund the difference if the 50% savings target is not realized. The firm thereby also internalizes all the risks normally borne by customers.

Increasing Interest in the Technology

The sample firms use a range of strategies to increase interest in the technology among their customers. First, at the onset of a project, firms targeting tenants have to exert considerable effort

Table 2. Description of Observed Compensating and Enhancing Tactics.

Compensating tactics	Enhancing tactics
Simplify: Take away or reduce the complexity and perceived risk of the offer for customers.	**Configure:** Develop modular offers to enable customers to configure the offer to match their needs more closely.
Mimic: Propose offers that resemble what customers are familiar with in the same or another industry to facilitate analogical reasoning.	**Engage:** Interact frequently with customers during product/service use to strengthen customer relations and possibly enable value cocreation.

to convince them to switch to electricity produced on-site. As one interviewee noted, electricity is a "low-interest" product and their tenants do not see the need for such offerings. Elderly people, especially, find these offerings too complex. As a result, the firms need to spend time explaining the concept and its added value to tenants. They may organize events and individual discussions to convince them to take part in an initiative and buy locally produced electricity. Once tenants sign delivery contracts, however, the three firms with Mieterstrom offerings only have limited interactions with customers beyond sending them regular billing statements. As such, none of these firms have adopted tactics for increasing interest in their technology. However, they need to make specific efforts to market their offerings.

Another strategy used by some of the sample firms involves enabling customers to customize their offerings. DZ-4, for instance, helps customers customize their PV systems to best serve their specific interests. This utility also allows people to lease either PV panels or PV panels combined with storage systems depending on the rate of self-sufficiency they would like to achieve. Moreover, customers have access to a smart phone app and a computer browser and can monitor their PV systems at any time.

Beegy goes even a step further and offers what they call a "learning" PV plant. Through an innovative online portal and using algorithm-based learning, Beegy provides weekly information on individual customers' consumption, production, self-consumption, cost savings, and predictions for the coming week. Customers also receive tailored feedback and advice on how to increase the rate of self-consumption. Beegy plans to offer additional services in the future to increase customer motivation. One of the plans envisions the creation of communities to enable customers to exchange electricity with each other and to further take ownership and control over their electricity use.

Four Tactics to Develop Attractive Value Propositions for Sustainable Technologies

Based on the results presented above, this section proposes four tactics that firms can use to develop attractive value propositions for sustainable technologies (see Table 2). Two tactics relate to and enrich the compensating tactics identified by Bohnsack and Pinkse (2017) and the other two relate to and enrich the enhancing tactics identified by the same authors.

Compensating Tactics

First, like Bohnsack and Pinkse (2017), we observed that firms adopt compensating tactics and try to turn what customers perceive as points of inferiority into points of parity with standard offers. One of the deterrents of PV technology is that it incurs high upfront costs. Some firms (DZ-4, Engynious, and HEG) therefore try to broaden their customer segments by offering PV

systems for lease, thereby making it possible for consumers to access PV technology without having to bear high upfront expenses. This strategy, however, has a significant impact on their business models as it determines the firms' corresponding revenue models (Tukker, 2004). By charging monthly fees, the firms secure recurring revenues, which provides some financial visibility. However, these asset heavy business models require securing large investments which may restrict the speed and extent to which firms can grow. Because in the digital age investors prefer firms with few physical assets (Libert, Beck, & Wind, 2016), these business models may be difficult to scale up.

In addition to these findings, two subcategories of compensating tactics emerged from our analysis: the simplifying tactic and the mimicking tactic.

Simplifying Tactics. First, all sample firms have developed at least one tactic for minimizing the complexity and perceived risk of the products or services proposed. Their objective is to make the customer experience as easy as possible by internalizing (part of) the technological complexity and technological risk, thereby also building trust in the offering. They help customers choose the best available technology, organize the installation and maintenance of the plant, and provide guarantees for the products purchased. Results indicate that this is in practice a key tactic for overcoming points of inferiority. Hassle-free offerings are expected to attract a larger customer segment. Arguably, simplifying tactics may be easier to implement for incumbents than for new entrants insofar as incumbents can leverage existing customer relations and their reputations to gain customer trust. Customers may more easily trust the technological products selected by an incumbent than those selected by a new entrant (Obal, 2013).

Regarding its impact on the business model, this tactic influences how firms construct the value networks that deliver value to customers. For example, RWE developed a partnership with AXA SA and DZ-4 has multiple partnerships with local installers. Here local conditions may facilitate or hinder the introduction of simplifying tactics. For instance, in Germany, many competent PV installers can be found throughout the country, which makes it possible to externalize this activity. In the United States, however, this is not the case. For example, SolarCity, one of the leading full-service solar power systems providers in the United States, has to employ its own installers (SolarCity, 2017). Moreover, internalizing the technological risk can also affect the firms' cost structures. By guaranteeing 50% savings on a customer's electricity bill, Beegy, for example, faces the risk of having to pay extra if those savings are not achieved.

Mimicking Tactics. We also observe that most firms (DZ-4, HEG, and Engynious Lichtblick), instead of selling products (e.g., PV with and without storage systems), prefer selling electricity supply contracts. In essence, this makes their value propositions very similar to what customers are accustomed to with standard electricity suppliers. We call this a mimicking tactic. By mimicking standard value propositions, firms facilitate customers' analogical reasoning by providing a familiar frame of reference (Gregan-Paxton & John, 1997), making it easier for customers to compare a new offering with a familiar one (Gregan-Paxton, Hibbard, Brunel, & Azar, 2002). Moreover, mimicking business models that are typical in other industries makes the new offering easier to comprehend (Gregan-Paxton & Moreau, 2003). For instance, because leasing models are very common in the car industry, customers can draw an analogy between a new PV offer and offerings they are already familiar with from other domains. Customers are thereby more likely to adopt the new technology (Martins, Rindova, & Greenbaum, 2015). Additional analogies may be found by looking at other sectors. Kamprath (2015), for instance, suggests looking at the creative industry and Mikhalkina and Cabantous (2015) recommend the sharing economy (see also Enkel & Mezger, 2013; Rumble & Minto, 2017).

Mimicking what is done in other domains implies identifying and adapting specific components of a business model (Enkel & Mezger, 2013). This can require a complete change of

mind-set. For example, in our case, PV panels are traditionally sold as products rather than as services. Making the switch to a service-oriented model deeply affects the business model as it defines the core of the value proposition, which becomes delivering a service instead of selling a product. Selling a service also influences the value chain needed to deliver the service. Indeed, many case study firms developed partnerships with green electricity suppliers to enable them to deliver residual power. Finally, being service-oriented also determines the revenue model and allows firms to generate recurring revenues.

Enhancing Tactics

Like Bohnsack and Pinkse (2017), we also observed that firms adopt various enhancing tactics to connect with untapped customer values. Our results enable us to further specify what enhancing tactics entail: they involve either a configuration tactic (when the product is sold) or an engaging tactic (during product use).

Configuration Tactics. When selling products (or services), some firms deployed *configuration* tactics. Two firms (RWE and DZ-4) allow customers to configure their offerings to suit their own individual wishes: they can choose how energy-independent they want to be, how much they would like to pay on a monthly basis or adapt an offering to their own housing situations. In this way, the firms develop business models that are hybrids of the "bus" (mass market) and "taxi" (project-based) business models observed in the literature (Baden-Fuller & Haefliger, 2013). By allowing for some modularity, firms are able to provide unique combinations to customers (Pine, 1993) and increase the variety of their offerings while maintaining the capacity to produce on a mass scale (Zipkin, 2001), enhancing the value of their offers by addressing several key customer requirements. Even though the market for self-sufficiency is still a small niche, it does require further refinement if it is to address the diversity of customers' needs and requirements that exist in this niche market (Da Silveira, Borenstein, & Fogliatto, 2001; Kotler, 1989; Pine, Victor, & Boynton, 1993).

This affects the business model in various ways. For instance, it requires firms to develop strategies for interacting with customers during the sales process to obtain the information required to develop an offering that is specifically tailored to customers' needs (Zipkin, 2001). It also demands the flexibility needed to assemble various combinations of products and services without generating high additional costs (Zipkin, 2001).

Engaging Tactics. Firms targeting homeowners also developed a range of tactics for *engaging with* customers when they are using products or services. We observed important differences depending on whether firms expect customers to be passively or actively involved with a product. The implications for the business models vary across firms. For instance, RWE offers regular monitoring services. The firm does not expect customers to be actively involved with their PV systems. A similar tactic is used by DZ-4. This tactic has only a minor impact on the business models of the firms and mostly influences how the firms capture value. By securing long-term customer relations, the firm potentially increases customer loyalty (Dick & Basu, 1994). In these times of deep transformations when customers are increasingly able to participate in the production of electricity, finding ways to retain customers becomes more and more important (PwC, 2016). Such enhancing tactics could enable incumbents to retain customers by gluing customers to firms and creating sticky relationships (Lemon, Rust, & Zeithaml, 2001).

Beegy adopted another strategy and developed an enhancing tactic based on the assumption that customers will be actively involved with their PV systems. By giving them regular and tailored feedback and advice based on their levels of consumption and production, Beegy empowers customers and enables them to take full control over their energy consumption. At the same

Table 3. Tactics Used by the Sample Firms to Develop Attractive Value Propositions for Power Supply Self-Sufficiency.

Reconfiguration tactics	Homeowners			Tenants		
	Beegy	DZ	RWE	Engynious	Lichtblick	HEG
Compensate						
Compensate		X		X		X
Simplify	X	X	X	X	X	X
Mimic		X		X	X	X
Enhance						
Configure	X	X	X			
Engage passive		X	X			
Engage cocreate	X					

time, Beegy also needs customers to take ownership of the systems to be able to deliver the promised (and guaranteed) service. As such, customers become cocreators of value (Prahalad & Ramaswamy, 2004): Beegy proposes helping customers become more self-sufficient by consuming electricity when it is being produced. This however requires customers to be flexible and to act on the feedback provided to them. This has several impacts on a firm's business model as it requires mastering digital technologies to be able to develop the appropriate interface allowing information to be communicated to customers and eventually triggering behavioral changes. To be successful, a firm should gain the trust of its customers and understand and leverage customer motivations, a process that typically takes time and requires multiple iterations (Gangale, Mengolini, & Onyeji, 2013). Even though this may be difficult to achieve, adopting such a strategy can enable a firm to identify currently untapped customer values and anticipate changes in customer demand. This may help firms reinvent their customer relationships, which is key for utilities seeking to gain full ownership of distributed energy generation (Schoettl & Lehman-Ortega, 2011).

Preliminary Conclusions

Table 3 shows the combinations of tactics that the firms in the case study employed to develop more attractive value propositions. Accordingly, two tactics are widely used by firms: simplifying and mimicking. The predominance of these tactics suggests that they are becoming key success factors in the commercialization of PV technology as most firms use them in one form or another to facilitate purchases. In the case of Germany, the predominance of simplifying tactics is a sign that the sector is gaining maturity. Firms no longer focus on purely technical aspects but consider the entire system surrounding the technology (e.g., contracts, installation, and maintenance). Similarly, most firms use mimicking tactics to develop value propositions that emphasize selling services (e.g., leasing or full electricity supply) instead of products. This points to a change of mind-set as firms move away from the more traditional product-based business models.

Moreover, Table 3 suggests that while firms targeting tenants use compensating tactics only, firms targeting homeowners mobilize a greater variety of tactics. This suggests that in the homeowner segment, firms strive to gain a competitive advantage through service differentiation. This observation also resonates with the conclusions drawn by Strupeit and Palm (2016).

Besides, two of the case study firms use innovative combinations of the aforementioned tactics. DZ-4 uses most of the tactics identified to increase the market attractiveness of their value proposition. Beegy uses both simplifying and engaging tactics through cocreation. This may seem counterintuitive at first sight. Indeed, these tactics decrease and increase, respectively, the

cognitive effort required from customers. However, while simplifying tactics aim to eliminate (part of) the technological and organizational complexity, engaging tactics through cocreation provide the technical infrastructure that empowers customers to take control over their energy consumption by using their current knowledge. This requires developing a customer interface that is easy to use, appealing and able to present information so that users can easily interpret and act on it. Beegy's value proposition relies heavily on the use of digital technologies which are quite new to the energy sector. In the following chapter, we will discuss and try to explain the differences across cases.

Comparing and Contrasting the Cases: A Cross Case Discussion

This section aims to compare and contrast the cases analyzed in order to explain similarities and difference observed between the cases regarding the combination of tactics they mobilize. The discussion will focus on explaining why some firms use more innovative combinations of tactics than other, what the combination of tactics used tell us about the type of customer interaction that is envisioned and the apparent limited possibilities to enhance value propositions for business models targeting tenants in multifamily houses.

Combining Tactics: Differences Across Companies

The findings suggest that the incumbent RWE adopts a rather traditional approach, while the two start-ups have developed more innovative business models. RWE uses a product-based business model and simplifies the customer purchase experience in a way that implies a rather low level of engagement with customers. In contrast, DZ-4 uses all the tactics that can help make solar electricity affordable, trustworthy, and as simple as possible for the customer and Beegy tries to find a balance between simplifying the customer experience and actively engaging customers to take control of their energy consumption (see Table 3).

Extant literature discusses why entrepreneurial firms are more likely to develop innovative business models than incumbents (Amit & Zott, 2001; Bohnsack et al., 2014; Chesbrough & Rosenbloom, 2002). It has been argued that entrepreneurial firms have the capacity to explore alternative business models based on novel ways to create value for customers (Chesbrough & Rosenbloom, 2002), and the flexibility required to pursue radically different business models (Sosna, Trevinyo-Rodríguez, & Velamuri, 2010). As such, they are more likely to develop value propositions based on an innovative combination of tactics. In comparison, incumbents tend to stay close to the existing business logic (Chesbrough & Rosenbloom, 2002), possibly because implementing a new business model that coexists with the old business model could lead to inefficiencies (Moingeon & Lehmann-Ortega, 2010). Besides, these explanations seem especially relevant for the energy sector where incumbents have been shown to be poorly equipped to develop business models for small-scale energy generation that require radical reimagining of the ways to engage customers (Richter, 2013). This suggests that incumbents may have the tendency to mobilize only those tactics that match their existing business logic.

It is however interesting to mention that a closer look at the company background shows that Beegy was cofounded by a consortium of firms including an incumbent regional utility.

Previous research argued that large regional energy companies can play a crucial role in the development of innovative products and services (Berlo &Wagner, 2013). Our findings support this argument. Incumbent regional utilities may be more likely to combine the agility of an entrepreneurial firm that is needed to mobilize engaging tactics while benefiting from the established market presence needed to penetrate mass markets (Hockerts & Wüstenhagen, 2010).

Finally, we observe that Beegy is the only firm in our sample that uses an engaging tactic through cocreation which is the most innovative tactic observed. As discussed previously, this

tactic requires mastering digital technologies which are new to the energy sector. Beegy is the result of a strategic partnership from firms coming from the energy and the information technology sector. Together, they codeveloped this innovative offer, each partner contributing its own knowledge and expertise (see also Svahn, Mathiassen, & Lindgren, 2017; von Stamm, 2004). This may explain why few companies are able to use this tactic.

Combining Tactics to Target Different Customer Segments

Electricity has long been a low-interest commodity with very homogeneous customer demand. The market for self-sufficiency allows imagining new ways to create value for customers and to interact with them. The results show that firms targeting homeowners have different visions about how they can develop attractive value propositions to best meet customer demands. Indeed, they focus on very different barriers to the adoption of PV technology and mobilize different combinations of tactics to increase the market attractiveness of their offer. Beegy simplifies customer experience while also engaging them to take robust ownership and places them at the heart of the energy system while DZ-4, and to a lesser extent RWE, focus on making solar electricity as simple as possible while engaging them on the margin. These differences can be revealing of the exploratory nature of companies' quests for new business models in what represents an emerging market for which there is no dominant business model yet (Morris, Schindehutte, & Allen, 2005). These differences can also reflect considerable uncertainties over future customer demand and point to a diversification of the customer segments depending on the level of engagement they are willing to show. Some firms have built their business models based on the assumption that electricity will remain a low-interest product, and that it is sufficient to offer products that are environmentally friendly and price competitive. Some expect that an increasing number of customers will be keen to have more control over how they can produce and consume electricity. In other words, the utilities segment their customers based on different expectations about customer's willingness to actively engage with their energy. Parag and Sovacool (2016) call for more research on how prosumers (consumers who also produce electricity) can be integrated more effectively and efficiently in the electricity markets. Our findings indicate that more effective integration will require better knowledge about who these prosumers are, and what they value. This knowledge may be acquired by mobilizing customers and integrating them in the business model (Plé, Lecocq, & Angot, 2010).

Since the time we collected our data, more companies have developed offers for self-sufficiency in the power supply in the German market. Similar to the cases we studied, these offerings foresee varying degrees of engagement on the part of their customers. On the one hand, a very engaging offering was made by Polarstern, for example. Polarstern proposes that customers trade surplus electricity with one another, thus interpreting customer demands in a way that is similar to Beegy. On the other hand, ESWE Versorgung, for example, has developed a value proposition which—similar to DZ-4—includes simple leasing offerings for installation, maintenance, and monitoring.

Limited Possibilities to Enhance Value Proposition for Tenant

Table 3 shows that firms that target tenants instead of homeowners, that is, Mieterstrom-type offerings, have developed rather similar value propositions. Moreover, none of the firms analyzed uses enhancing tactics. Our findings also show that customer relations are long-term and intensive in the homeowner segment. In contrast, they are short-term and rather weak in the tenant segment. Two lines of argument may explain these findings. First, Mieterstrom offerings are technically complex to set up. They require very specific metering and billing processes

which make it possible to differentiate between tenants who consume locally and those who do not, and between the portion of electricity consumed that is produced locally and the portion that is bought from the grid. Moreover, Mieterstrom offerings also require close working relationships with DSOs, which often are unfamiliar with these new offerings. At the time of data collection, very few pilot projects had been developed. Thus, experience was scarce. These technical issues might contribute to explaining why the firms have not yet differentiated their offers. Second, firms targeting homeowners can focus on a specific market segment and develop offerings that are more accurately tailored to what they need and value. In contrast, with Mieterstrom offerings, firms can target end users only indirectly, since these typically are chosen as tenants by intermediaries (property owners). These tenants tend to be rather heterogeneous and include students, young families with children, and retirees. Since the preferences of such heterogeneous tenants differ, utilities might find it difficult to develop tailored offerings that engage customers.

Since the time we gathered our data, the Renewable Energy Act has been amended (on 25 July 2017) to increase the economic profitability of Mieterstrom offerings. These offerings now have the right to claim a special subsidy meant to compensate for the higher costs incurred in billing and metering activities. This has stimulated more firms to enter this market. However, even though the market potential is large—in Germany, the rate of home ownership is only around 52%, well below the EU average of 70% (Eurostat, 2018)—the market has developed rather slowly (Engelke, 2018). This might be because Mieterstrom offerings are still too complicated to set up for firms and that regulations fail to create attractive market conditions for these types of offers. Besides, according to our interviews, the connection rate, that is, the percentage of tenants who decide to switch to locally supplied electricity, typically ranges between only 10% and 25% for a particular project. Firms proposing Mieterstrom offerings for multifamily rental houses may also need to find ways to mobilize tactics to enhance their value propositions if they want to increase this connection rate, generate more profit from these types of offers, and reach larger parts of the market.

Conclusions

Making the transition to sustainable development requires the sustainable transformation of mass markets (Schaltegger et al., 2016). However, sustainable technologies often lack market attractiveness (Johnson & Suskewicz, 2009). This article explores how firms can develop compelling value propositions to increase the market attractiveness of sustainable technologies that are mature and competitive in terms of price but are difficult to move beyond niche markets. Building on and refining the work of Bohnsack and Pinkse (2017) for disruptive technologies, this article presents four tactics that firms can use to overcome the perceived inferiorities of sustainable technologies: simplifying, mimicking, configuring, and engaging. Even though these tactics have been derived from the analysis of a single technology, namely PV technology and more specifically the market for self-sufficiency in power supply, we believe the findings to be useful for other sustainable technologies as well. Indeed, sustainable technologies often face barriers to adoption that are not directly linked with their technological or economic performance but with sociocultural or organizational aspects (see, for instance, Anttonen et al., 2013; Hoffman & Henn, 2008; Sousa-Zomer & Cauchick Miguel, 2016). These perceived inferiorities cannot be solved by technological innovations but could be addressed by reconfiguring the value proposition in a way that helps reduce these barriers.

This article provides some guidance to firms seeking to increase the market attractiveness of sustainable technologies and to overcome customer inertia. First, it is crucial to identify the barriers hindering the adoption of a given sustainable technology to understand why customers perceive the technology as inferior. Second, firms can use the tactics identified to reconfigure

their value proposition so as to reduce these barriers. The tactics presented in this article invite firms to consider their market differently and view barriers to adoption as opportunities to develop innovative and differentiating business models. The tactics presented can also help firms imagine new ways to interact with customers. This may be particularly important for sustainable technologies that are developed in sectors that have traditionally featured a low level of customer engagement, such as heating, energy efficiency, (waste) water, and gas. Our article also suggests that some combinations of tactics are more innovative than others. Firms may need to engage in strategic partnerships in order to gain access to complementary assets (e.g., network of installers) or to new competences (e.g., digital competences).

We conclude by noting that there is growing interest in studying business models for sustainable technologies and we believe there are numerous stimulating avenues for future research. This study analyzed the value propositions of firms operating in a single country and it would be interesting to assess whether and to what extent the tactics mobilized depend on the cultural and political context where the business models have been coined. Finally, since the time we collected data, the EEG has gone through series of amendment, influencing the size and attractiveness of the market. A longitudinal study could analyze how this has influenced the way firms construct their value proposition. Such study would help better grasp the role of different regulatory environments in shaping firm's business model.

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Notes

1. Feed-in-tariffs are fixed payments to electricity generators for each kWh of electricity supplied from renewable energy sources.
2. Being self-sufficient means that households produce as much electricity as they consume. For example, combining rooftop PV and battery storage systems may currently achieve self-sufficiency rates of up to 70%. If more electricity is generated than can be consumed it is fed back into grid and receives a feed-in-tariff (currently ca. 11 cts/kWh in Germany). If more electricity is used than can be generated, it is possible to get residual loads from the grid (by paying currently ca. 29 cts/kWh in Germany). This also shows that it is more economically attractive to self-consume than to feed electricity to the grid.
3. Grid parity reflects the capacity of alternative energy sources such as PV or wind turbines to generate power at a levelized cost of electricity below or equal to the price of purchasing electricity from the grid.
4. This increase may be explained also by regulatory changes. Indeed, since 2012, German regulation requires that operators of installations exceeding 10 KWp either self-consume or market externally at least 10% of the electricity generated.

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References

Amit, R., & Zott, C. (2001). Value creation in E-business. *Strategic Management Journal*, *22*, 493-520. doi:10.1002/smj.187

Anderson, J. C., Narus, J. A., & Van Rossum, W. (2006, March). Customer value propositions in business markets. *Harvard Business Review*. Retrieved from https://hbr.org/2006/03/customer-value-propositions-in-business-markets

Anttonen, M., Halme, M., Houtbeckers, E., & Nurkka, J. (2013). The other side of sustainable innovation: Is there a demand for innovative services? *Journal of Cleaner Production*, *45*, 89-103. doi:10.1016/j.jclepro.2011.12.019

Asmus, P. (2008). Exploring new models of solar energy development. *Electricity Journal*, *21*, 61-70. doi:10.1016/j.tej.2008.03.005

Auer, H., & Lettner, G. (2012). *Realistic roadmap to PV grid parity for all target countries*. Vienna, Austria: Vienna University of Technology.

Baden-Fuller, C., & Haefliger, S. (2013). Business models and technological innovation. *Long Range Planning*, *46*, 419-426. doi:10.1016/j.lrp.2013.08.023

Berlo, K., & Wagner, O. (2013). *Stadtwerke-Neugründungen und Rekommunalisierungen: Energieversorgung in kommunaler Verantwortung: Bewertung der 10 wichtigsten Ziele und deren Erreichbarkeit* [Utilities start-ups and re-municipalisations: Energy supply in municipal responsibility: evaluation of the 10 most important goals and their achievability]. Retrieved from https://www.energie-experten.org/uploads/media/Studie_Stadtwerke-Neugr%C3%BCndungen_und_Rekommunalisierungen.pdf

Blomberg New Energy Finance. (2015). *New Energy Outlook 2015: Long term projection of the global energy sector*. New York, NY: Bloomberg Finance.

BMUB & BMWi. (2011). *The Federal Government's energy concept of 2010 and the transformation of the energy system of 2011*. Berlin, Germany: Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety and Federal Ministry for Economic Affairs and Energy.

Bocken, N. M. P., de Pauw, I., Bakker, C., & van der Grinten, B. (2016). Product design and business model strategies for a circular economy. *Journal of Industrial and Production Engineering*, *33*, 308-320. doi:10.1080/21681015.2016.1172124

Bocken, N. M. P., & Short, S. W. (2016). Towards a sufficiency-driven business model: Experiences and opportunities. *Environmental Innovation and Societal Transitions*, *18*, 41-61. doi:10.1016/j.eist.2015.07.010

Bocken, N. M. P., Short, S. W., Rana, P., & Evans, S. (2014). A literature and practice review to develop sustainable business model archetypes. *Journal of Cleaner Production*, *65*, 42-56. doi:10.1016/j.jclepro.2013.11.039

Bohnsack, R., & Pinkse, J. (2017). Value propositions for disruptive technologies: Reconfiguration tactics in the case of electric vehicles. *California Management Review*, *59*, 79-96. doi:10.1177/0008125617717711

Bohnsack, R., Pinkse, J., & Kolk, A. (2014). Business models for sustainable technologies: Exploring business model evolution in the case of electric vehicles. *Research Policy*, *43*, 284-300. doi:10.1016/j.respol.2013.10.014

Boons, F., & Lüdeke-Freund, F. (2013). Business models for sustainable innovation: State-of-the-art and steps towards a research agenda. *Journal of Cleaner Production*, *45*, 9-19. doi:10.1016/j.jclepro.2012.07.007

CE Delft. (2016). *The potential of energy citizens in the European Union*. Retrieved from http://www.cedelft.eu/publicatie/the_potential_of_energy_citizens_in_the_european_union/1845

Charter, M., Gray, C., Clark, T., & Woolman, T. (2008). Review: The role of business in realising sustainable consumption and production. In A. Tukker, M. Charter, C. Vezzoli, E. Stø, & M. M. Anderson (Eds.), *Perspectives on radical changes to sustainable consumption and production 1: System innovation for sustainability* (pp. 46-69). Sheffield, England: Greenleaf.

Chesbrough, H., & Rosenbloom, R. S. (2002). The role of the business model in capturing value from innovation: Evidence from Xerox Corporation's technology spin-off companies. *Industrial and Corporate Change*, *11*, 529-555. doi:10.1093/icc/11.3.529

Christensen, C. M. (1997). *The innovator's dilemma: When new technologies cause great firms to fail*. Boston, MA: Harvard Business School Press.

Cooremans, C. (2011). Make it strategic! Financial investment logic is not enough. *Energy Efficiency*, *4*, 473-492. doi:10.1007/s12053-011-9125-7

Da Silveira, G., Borenstein, D., & Fogliatto, F. S. (2001). Mass customization: Literature review and research directions. *International Journal of Production Economics*, *72*(1), 1-13. doi:10.1016/S0925-5273(00)00079-7

del Río, P., & Bleda, M. (2012). Comparing the innovation effects of support schemes for renewable electricity technologies: A function of innovation approach. *Energy Policy*, *50*, 272-282. doi:10.1016/j.enpol.2012.07.014

Dick, A. S., & Basu, K. (1994). Customer loyalty: Toward an integrated conceptual framework. *Journal of the Academy of Marketing Science*, *22*, 99-113. doi:10.1177/0092070394222001

Drury, E., Miller, M., Macal, C. M., Graziano, D. J., Heimiller, D., Ozik, J., & Perry, T. D., IV. (2012). The transformation of southern California's residential photovoltaics market through third-party ownership. *Energy Policy*, *42*, 681-690. doi:10.1016/j.enpol.2011.12.047

Engelke, T. (2018). Ein Jahr Mieterstromgesetz [A year tenant Electricity Act]. Berlin, Germany: Verbraucherzentrale Bundesverband.

Engynious Energy Solutions. (2018). *Engynious Joint Venture with Younergy completes 200th PV plant: Energy Communities as future new segment*. Retrieved from https://www.engynious.com/en/news/news/engynious-joint-venture-200/

Enkel, E., & Mezger, F. (2013). Imitation processes and their application for business model innovation: An explorative study. *International Journal of Innovation Management*, *17*(01), 1340005. doi:10.1142/S1363919613400057

European Commission. (2014). *PV Status Report 2014*. Retrieved from https://setis.ec.europa.eu/sites/default/files/reports/PV-status-report-2014.pdf

Eurostat. (2018). *Housing statistics*. Retrieved from https://ec.europa.eu/eurostat/statistics-explained/index.php/Housing_statistics

Fischer, C. (2008). Feedback on household electricity consumption: A tool for saving energy? *Energy Efficiency*, *1*, 79-104. doi:10.1007/s12053-008-9009-7

Fleiter, T., Schleich, J., & Ravivanpong, P. (2012). Adoption of energy-efficiency measures in SMEs: An empirical analysis based on energy audit data from Germany. *Energy Policy*, *51*, 863-875. doi:10.1016/j.enpol.2012.09.041

Foxon, T., & Pearson, P. (2008). Overcoming barriers to innovation and diffusion of cleaner technologies: Some features of a sustainable innovation policy regime. *Journal of Cleaner Production*, *16*(1, Suppl. 1), S148-S161. doi:10.1016/j.jclepro.2007.10.011

Foxon, T. J. (2002, September). *Technological and institutional "lock-in" as a barrier to sustainable innovation*. Paper presented at the International Summer School on Innovation for Sustainable Development: Institutions, Incentives and Economic Policy, Seeon, Germany. Retrieved from https://www.imperial.ac.uk/media/imperial-college/research-centres-and-groups/icept/7294726.PDF

Fraunhofer Institute for Solar Energy Systems ISE. (2017). *Power generation in Germany: Assessment of 2016*. Retrieved from https://www.ise.fraunhofer.de/content/dam/ise/en/documents/publications/studies/power-generation-from-renewable-energies-2016.pdf

Funkhouser, E., Blackburn, G., Magee, C., & Rai, V. (2015). Business model innovations for deploying distributed generation: The emerging landscape of community solar in the U.S. *Energy Research & Social Science*, *10*, 90-101. doi:10.1016/j.erss.2015.07.004

Gangale, F., Mengolini, A., & Onyeji, I. (2013). Consumer engagement: An insight from smart grid projects in Europe. *Energy Policy*, *60*, 621-628. doi:10.1016/j.enpol.2013.05.031

Gillingham, K., & Palmer, K. (2014). Bridging the energy efficiency gap: Policy insights from economic theory and empirical evidence. *Review of Environmental Economics and Policy*, *8*, 18-38. doi:10.1093/reep/ret021

Gregan-Paxton, J., Hibbard, J. D., Brunel, F. F., & Azar, P. (2002). "So that's what that is": Examining the impact of analogy on consumers' knowledge development for really new products. *Psychology & Marketing*, *19*, 533-550. doi:10.1002/mar.10023

Gregan-Paxton, J., & John, D. R. (1997). Consumer learning by analogy: A model of internal knowledge transfer. *Journal of Consumer Research*, *24*, 266-284. doi:10.1086/209509

Gregan-Paxton, J., & Moreau, P. (2003). How do consumers transfer existing knowledge? A comparison of analogy and categorization effects. *Journal of Consumer Psychology*, *13*, 422-430. doi:10.1207/S15327663JCP1304_09

Heidelberger Energiegenossenschaft. (2018a). *Mehrfamilienhausmodell: Workshops und Wissen* [Multi-family house model: workshops and knowledge]. Retrieved from https://www.heidelberger-energiegenossenschaft.de/projekte/weitere-projekte/mehrfamilienhaus-modell

Heidelberger Energiegenossenschaft. (2018b). *Unternehmensgeschichte* [Company History]. Retrieved from https://www.heidelberger-energiegenossenschaft.de/ueber-uns/unternehmensgeschichte

Henkel, J., & Lenck, T. (2013). *Prognose der Stromabgabe an Letztverbraucher bis 2018: Gutachten für die vier deutschen Übertragungsnetzbetreiber im Auftrag der TransnetBW GmbH* [Forecast of electricity supply to final consumers by 2018: Report for the four German transmission system operators on behalf of TransnetBW GmbH]. Retrieved from https://www.energybrainpool.com/fileadmin/download/Studien/Studie_2013-10-27_Prognose_LV_MiFri_Bericht_EnergyBrainpool.pdf

Hockerts, K., & Wüstenhagen, R. (2010). Greening Goliaths versus emerging Davids: Theorizing about the role of incumbents and new entrants in sustainable entrepreneurship. *Journal of Business Venturing*, *25*, 481-492. doi:10.1016/j.jbusvent.2009.07.005

Hoffman, A. J., & Henn, R. (2008). Overcoming the social and psychological barriers to green building. *Organization & Environment*, *21*, 390-419. doi:10.1177/1086026608326129

Hoppmann, J., Huenteler, J., & Girod, B. (2014). Compulsive policy-making: The evolution of the German feed-in tariff system for solar photovoltaic power. *Research Policy*, *43*, 1422-1441. doi:10.1016/j.respol.2014.01.014

Howard-Grenville, J. A., & Hoffman, A. J. (2003). The importance of cultural framing to the success of social initiatives in business. *Academy of Management Executive*, *17*, 70-86.

IEA-RETD. (2016). *RE-TRANSITION: Transitioning to policy frameworks for cost-competitive renewables*. Retrieved from http://iea-retd.org/wp-content/uploads/2016/03/IEA-RETD_RE-TRANSITION.pdf

Jaffe, A. B., & Stavins, R. N. (1994). The energy-efficiency gap: What does it mean? *Energy Policy*, *22*, 804-810. doi:10.1016/0301-4215(94)90138-4

Johnson, M. W., & Suskewicz, J. (2009, November). How to jump-start the clean-tech economy. *Harvard Business Review*. Retrieved from https://hbr.org/2009/11/how-to-jump-start-the-clean-tech-economy

Kamprath, M. (2015). *A microfoundations perspective on foresight and business models: Competence development and business model design in times of digital industry transformation* (Master thesis). University of Potsdam, Berlin, Germany. Retrieved from https://www.researchgate.net/publication/277019283_A_microfoundations_perspective_on_foresight_and_business_models_-_Competence_development_and_business_model_design_in_times_of_digital_industry_transformation

Karakaya, E., Hidalgo, A., & Nuur, C. (2015). Motivators for adoption of photovoltaic systems at grid parity: A case study from Southern Germany. *Renewable and Sustainable Energy Reviews*, *43*, 1090-1098. doi:10.1016/j.rser.2014.11.077

Kley, F., Lerch, C., & Dallinger, D. (2011). New business models for electric cars: A holistic approach. *Energy Policy*, *39*, 3392-3403. doi:10.1016/j.enpol.2011.03.036

Kotler, P. (1989). From mass marketing to mass customization. *Planning Review*, *17*(5), 10-47. doi:10.1108/eb054267

Lemon, K. N., Rust, R. T., & Zeithaml, V. A. (2001). What drives customer equity. *Marketing Management*, *10*(1), 20-25.

Libert, B., Beck, M., & Wind, Y. (Jerry). (2016, September 29). Investors today prefer companies with fewer physical assets. *Harvard Business Review*. Retrieved from https://hbr.org/2016/09/investors-today-prefer-companies-with-fewer-physical-assets

Lindič, J., & Marques da Silva, C. (2011). Value proposition as a catalyst for a customer focused innovation. *Management Decision*, *49*, 1694-1708. doi:10.1108/00251741111183834

Luthra, S., Kumar, S., Garg, D., & Haleem, A. (2015). Barriers to renewable/sustainable energy technologies adoption: Indian perspective. *Renewable and Sustainable Energy Reviews*, *41*, 762-776. doi:10.1016/j.rser.2014.08.077

Mankad, A., & Tapsuwan, S. (2011). Review of socio-economic drivers of community acceptance and adoption of decentralised water systems. *Journal of Environmental Management*, *92*, 380-391. doi:10.1016/j.jenvman.2010.10.037

Martins, L. L., Rindova, V. P., & Greenbaum, B. E. (2015). Unlocking the hidden value of concepts: A cognitive approach to business model innovation. *Strategic Entrepreneurship Journal*, *9*, 99-117. doi:10.1002/sej.1191

Mikhalkina, T., & Cabantous, L. (2015). Business model innovation: How iconic business models emerge. In C. Baden-Fuller & V. Mangematin (Ed.), *Business models and modelling* (Advances in Strategic Management, *Vol. 33*, pp. 59-95). Bingley, England: Emerald.

Moingeon, B., & Lehmann-Ortega, L. (2010). Creation and implementation of a new business model: A disarming case study. *M@n@gement*, *13*, 266-297. doi:10.3917/mana.134.0266

Moore, G. A. (1999). *Crossing the chasm: Marketing and selling high-tech products to mainstream customers*. New York, NY: Harper Business.

Morris, M., Schindehutte, M., & Allen, J. (2005). The entrepreneur's business model: Toward a unified perspective. *Journal of Business Research*, *58*, 726-735. doi:10.1016/j.jbusres.2003.11.001

Moshövel, J., Kairies, K.-P., Magnor, D., Leuthold, M., Bost, M., Gährs, S., . . . Sauer, D. U. (2015). Analysis of the maximal possible grid relief from PV-peak-power impacts by using storage systems for increased self-consumption. *Applied Energy*, *137*, 567-575. doi:10.1016/j.apenergy.2014.07.021

Negro, S. O., Alkemade, F., & Hekkert, M. P. (2012). Why does renewable energy diffuse so slowly? A review of innovation system problems. *Renewable and Sustainable Energy Reviews*, *16*, 3836-3846. doi:10.1016/j.rser.2012.03.043

Noack, M. (2018). *Solaranlage einfach mieten mit DZ–4* [Simply renting a solar system with DZ-4]. Retrieved from https://gruene-startups.de/solaranlage-einfach-mieten-mit-dz-4/

Obal, M. (2013). Why do incumbents sometimes succeed? Investigating the role of interorganizational trust on the adoption of disruptive technology. *Industrial Marketing Management*, *42*, 900-908. doi:10.1016/j.indmarman.2013.05.017

Olsthoorn, M., Schleich, J., & Hirzel, S. (2017). Adoption of energy efficiency measures for non-residential buildings: Technological and organizational heterogeneity in the trade, commerce and services sector. *Ecological Economics*, *136*, 240-254. doi:10.1016/j.ecolecon.2017.02.022

Parag, Y., & Sovacool, B. K. (2016). Electricity market design for the prosumer era. *Nature Energy*, *1*(4), 16032. doi:10.1038/nenergy.2016.32

Pine, B. J. (1993). Making mass customization happen: Strategies for the new competitive realities. *Planning Review*, *21*(5), 23-24. doi:10.1108/eb054435

Pine, B. J., Victor, B., & Boynton, A. C. (1993, September-October). Making mass customization work. *Harvard Business Review*. Retrieved from https://hbr.org/1993/09/making-mass-customization-work

Plé, L., Lecocq, X., & Angot, J. (2010). Customer-integrated business models: A theoretical framework. *M@n@gement*, *13*, 226-265.

Prahalad, C. K., & Ramaswamy, V. (2004). Co-creating unique value with customers. *Strategy & Leadership*, *32*(3), 4-9. doi:10.1108/10878570410699249

Prognos. (2016). *Eigenversorgung aus Solaranlagen: Das Potenzial für Photovoltaik-Speicher-Systeme in Ein- und Zweifamilienhäusern, Landwirtschaft sowie im Lebensmittelhandel* [Self-sufficiency with solar systems: the potential for photovoltaic storage systems in detached and semi-detached houses, agriculture and in the food trade sector]. Retrieved from https://www.agora-energiewende.de/fileadmin2/Projekte/2016/Dezentralitaet/Agora_Eigenversorgung_PV_web-02.pdf

PwC. (2016). *Customer engagement in an era of energy transformation*. Retrieved from https://www.pwc.nl/nl/assets/documents/pwc-customer-engagement-in-an-era-of-energy-transformation.pdf

REN21. (2016). *Renewables 2016 Global Status Report*. Retrieved from http://www.ren21.net/status-of-renewables/global-status-report/

Richter, M. (2013). Business model innovation for sustainable energy: German utilities and renewable energy. *Energy Policy*, *62*, 1226-1237. doi:10.1016/j.enpol.2013.05.038

Rosoff, L., & Sinclair, M. (2009). *Smart solar marketing strategies*. Retrieved from https://www.cesa.org/assets/Uploads/Resources-pre-8-16/CEG-Solar-Marketing-Report-2009.pdf

Rumble, R., & Minto, N. A. (2017). How to use analogies for creative business modelling. *Journal of Business Strategy*, *38*, 76-82. doi:10.1108/JBS-09-2016-0091

Sauter, R., & Watson, J. (2007). Strategies for the deployment of micro-generation: Implications for social acceptance. *Energy Policy*, *35*, 2770-2779. doi:10.1016/j.enpol.2006.12.006

Schaltegger, S., Lüdeke-Freund, F., & Hansen, E. G. (2012). Business cases for sustainability: The role of business model innovation for corporate sustainability. *International Journal of Innovation and Sustainable Development*, *6*, 95-119.

Schaltegger, S., Lüdeke-Freund, F., & Hansen, E. G. (2016). Business models for sustainability: A co-evolutionary analysis of sustainable entrepreneurship, innovation, and transformation. *Organization & Environment*, *29*, 264-289. doi:10.1177/1086026616633272

Schleicher-Tappeser, R. (2012). How renewables will change electricity markets in the next five years. *Energy Policy*, *48*, 64-75. doi:10.1016/j.enpol.2012.04.042

Schoettl, J., & Lehman-Ortega, K. (2011). Photovoltaic business model: Threat or opportunity for utilities? In R. Wüstenhagen & R. Wuebker (Eds.), *Handbook of research on energy entrepreneurship* (pp. 145-164). Cheltenham, England: Edward Elgar.

Shih, L. H., & Chou, T. Y. (2011). Customer concerns about uncertainty and willingness to pay in leasing solar power systems. *International Journal of Environmental Science and Technology*, *8*, 523-532. doi:10.1007/BF03326238

SolarCity. (2017). *SolarCity careers: Better energy starts with you*. Retrieved from http://www.solarcity.com/careers/

Sosna, M., Trevinyo-Rodríguez, R. N., & Velamuri, S. R. (2010). Business model innovation through trial-and-error learning: The Naturhouse case. *Long Range Planning*, *43*, 383-407. doi:10.1016/j.lrp.2010.02.003

Sousa-Zomer, T. T., & Cauchick Miguel, P. A. (2016). Sustainable business models as an innovation strategy in the water sector: An empirical investigation of a sustainable product-service system. *Journal of Cleaner Production*, *171*(Suppl.), S119-S129. doi:10.1016/j.jclepro.2016.07.063

Strupeit, L., & Palm, A. (2016). Overcoming barriers to renewable energy diffusion: Business models for customer-sited solar photovoltaics in Japan, Germany and the United States. *Journal of Cleaner Production*, *123*, 124-136. doi:10.1016/j.jclepro.2015.06.120

Svahn, F., Mathiassen, L., & Lindgren, R. (2017). Embracing digital innovation in incumbent firms: How Volvo cars managed competing concerns. *MIS Quarterly*, *41*, 239-254.

Teece, D. J. (2009). *Dynamic capabilities and strategic management: Organizing for innovation and growth*. Oxford, England: Oxford University Press.

Teece, D. J. (2010). Business models, business strategy and innovation. *Long Range Planning*, *43*, 172-194. doi:10.1016/j.lrp.2009.07.003

Tongur, S., & Engwall, M. (2014). The business model dilemma of technology shifts. *Technovation*, *34*, 525-535. doi:10.1016/j.technovation.2014.02.006

Trendresearch. (2013). *Anteile einzelner Marktakteure an Erneuerbare Energien-Anlagen in Deutschland* [Shares of individual market players in renewable energy systems in Germany] (2nd ed.). Bremen, Germany.

Tukker, A. (2004). Eight types of product–service system: Eight ways to sustainability? Experiences from SusProNet. *Business Strategy and the Environment*, *13*, 246-260. doi:10.1002/bse.414

Tukker, A., & Tischner, U. (2006). *New business for old Europe: Product-service development, competitiveness and sustainability*. Sheffield, England: Greenleaf.

Unruh, G. C. (2000). Understanding carbon lock-in. *Energy Policy*, *28*, 817-830. doi:10.1016/S0301-4215(00)00070-7

Unruh, G. C. (2002). Escaping carbon lock-in. *Energy Policy*, *30*, 317-325.

U.S. Department of Energy. (2012). *Photovoltaic (PV) pricing trends: Historical, recent, and near-term projections*. Retrieved from https://www.nrel.gov/docs/fy13osti/56776.pdf

Veugelers, R. (2012). Which policy instruments to induce clean innovating? *Research Policy*, *41*, 1770-1778. doi:10.1016/j.respol.2012.06.012

von Stamm, B. (2004). Collaboration with other firms and customers: Innovation's secret weapon. *Strategy & Leadership*, *32*(3), 16-20. doi:10.1108/10878510410535727

Wassermann, S., Reeg, M., & Nienhaus, K. (2015). Current challenges of Germany's energy transition project and competing strategies of challengers and incumbents: The case of direct marketing of electricity from renewable energy sources. *Energy Policy*, *76*, 66-75. doi:10.1016/j.enpol.2014.10.013

Wells, P. (2008). Alternative business models for a sustainable automotive industry. In A. Tukker, M. Charter, C. Vezzoli, E. Stø, & M. M. Anderson (Eds.), *Perspectives on radical changes to sustainable consumption and production 1: System innovation for sustainability* (pp. 80-98). Sheffield, England: Greenleaf.

Wells, P. (2016). Degrowth and techno-business model innovation: The case of Riversimple. *Journal of Cleaner Production*, *197*, 1704-1710. doi:10.1016/j.jclepro.2016.06.186

Yang, C.-J. (2010). Reconsidering solar grid parity. *Energy Policy*, *38*, 3270-3273. doi:10.1016/j.enpol.2010.03.013

Yu, D., & Hang, C. C. (2010). A reflective review of disruptive innovation theory. *International Journal of Management Reviews*, *12*, 435-452. doi:10.1111/j.1468-2370.2009.00272.x

Zhang, X., Shen, L., Tam, V. W. Y., & Lee, W. W. Y. (2012). Barriers to implement extensive green roof systems: A Hong Kong study. *Renewable and Sustainable Energy Reviews*, *16*, 314-319. doi:10.1016/j.rser.2011.07.157

Zipkin, P. (2001, April 15). The limits of mass customization. *MIT Sloan Management Review*. Retrieved from https://sloanreview.mit.edu/article/the-limits-of-mass-customization/

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